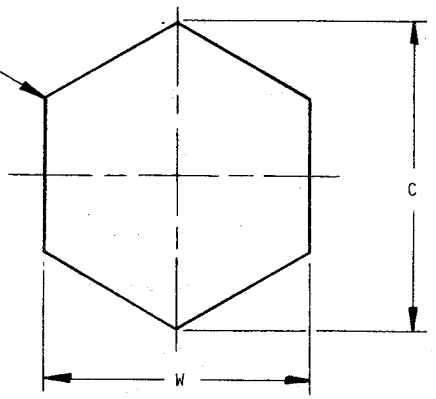
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FED. SUP CLASS
53GP



THIS STANDARD ESTABLISHES DIMENSIONS OF STANDARD HEXAGONS FOR FASTENERS

NOMINAL ACROSS FLATS	W BASIC SIZE ACROSS FLATS	W LIMITS OF SIZE ACROSS FLATS MIN[a]	W LIMITS OF SIZE ACROSS FLATS (A) MAX [b]	C ACROSS CORNERS MINIMUM [c][d]
5/32	.1562	.150	.1562	.171
* 3/16	.1875	.181	.1880	.207
7/32	.2188	.212	.2190	.242
(A) 1/4	.2500	.243	.2500	.277
9/32	.2812	.274	.2812	.313
* 5/16	.3125	.305	.3140	.348
*(A) 11/32	.3438	.336	.3450	.383
(A) 3/8	.3750	.367	.3760	.410
* 7/16	.4375	.429	.4380	.480
* 1/2	.5000	.492	.5020	.552
* 9/16	.5625	.554	.5640	.623
5/8	.6250	.616	.6270	.694
* 11/16	.6875	.678	.6900	.764
* 3/4	.7500	.741	.7530	.836
13/16	.8125	.803	.8160	.907
* 7/8	.8750	.865	.8780	.978
* 15/16	.9375	.928	.9420	1.050
1	1.0000	.990	1.0040	1.120
(A) 1-1/16	1.0625	1.053	1.0650	1.192
*(A) 1-3/16	1.1875	1.177	1.1910	1.346
1-1/8	1.1250	1.115	1.1290	1.263
1-3/16	1.1875	1.177	1.1910	1.334
(A) 1-1/4	1.2500	1.239	1.2540	1.405
1-5/16	1.3125	1.302	1.3170	1.476
(A) 1-7/16	1.4375	1.427	1.4430	1.619
* 1-1/2	1.5000	1.489	1.5050	1.690
* 1-5/8	1.6250	1.614	1.6310	1.832
*(A) 1-11/16	1.6875	1.676	1.6930	1.903
* 1-13/16	1.8125	1.801	1.8190	2.046
*(A) 1-7/8	1.8750	1.863	1.8820	2.117
2	2.0000	1.988	2.0080	2.259
*(A) 2-1/16	2.0625	2.050	2.0710	2.330
*(A) 2-1/8	2.125	2.113	(f) 2.1270	2.402
*(A) 2-1/4	2.250	2.238	2.2590	2.545
2-3/16	2.1875	2.175	2.1970	2.473
2-3/8	2.3750	2.360	2.3850	2.684

THIS IS A DESIGN STANDARD AND NOT TO BE USED AS A PART NO.

(A) REVISED AND REDRAWN

APPROVED 30 NOV 73
REVISED (A) 25 MAR 87

AMSC N/A

P.A.	INTERNATIONAL INTEREST	TITLE	MILITARY STANDARD
NAVY — AS Other Cust USAF - 11 ARMY - AV	*	WRENCHING ELEMENT, EXTERNAL HEXAGON, FOR THREADED FASTENERS	MS21132
PROCUREMENT SPECIFICATION NONE		SUPERSEDES:	SHEET 1 OF 2

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(A) (a) MINIMUM DIMENSION ACROSS FLATS = W -0.01 $\sqrt[4]{W}$ WHERE "W" IS THE BASIC SIZE EXPRESSED AS THE COMPLETE DECIMAL EQUIVALENT OF THE FRACTIONAL SIZE.

(b) THE ABOVE DIMENSIONS ARE COMPATIBLE WITH FED-STD-346 GAGES, WRENCH OPENINGS.

(c) FOR SIZES SMALLER THAN 3/8 (.375), DIMENSIONS ACROSS CORNERS C MINIMUM = (1.141) X (W MINIMUM).

(d) FOR SIZES 3/8 (.375) AND LARGER, DIMENSIONS ACROSS CORNERS C MINIMUM = (1.141) X (W MINIMUM) -(.009).

(e) THE HEXAGON SIZES LISTED HERE ARE BASED ON THE FOLLOWING CRITERIA:

1. THE SIZE REQUIRED TO ENVELOP THE OUTSIDE DIAMETER OF THE MINIMUM BEARING AREA OF THE BOLT HEAD AND NUT FACE REQUIRED BY THE BEARING YIELD STRENGTH OF THE BOLTED MATERIAL.
2. BOLTED MATERIAL BEARING YIELD STRENGTH OF 83 KSI.
3. TENSION BOLT MATERIAL TENSILE STRENGTH OF 125 KSI.
4. SHEAR BOLT MATERIAL ULTIMATE TENSILE STRENGTH OF 160 KSI.
5. SIZES FOR WHICH WRENCHES ARE AVAILABLE.
6. SIZES ARE NOT BASED ON THE DESIGN REQUIREMENTS OF FLANGED OR RING BASE BOLT HEADS OR NUTS.

(f) MAXIMUM DIMENSION ACROSS FLATS = W + 0.002.

* CERTAIN PROVISIONS (LIMITS FOR ACROSS FLATS DIMENSIONS (W)) OF THIS STANDARD ARE THE SUBJECT OF INTERNATIONAL STANDARDIZATION AGREEMENT ASCC 17/2 AIR STD. WHEN REVISION, OR CANCELLATION OF THIS STANDARD IS PROPOSED WHICH WILL AFFECT OR VIOLATE THE INTERNATIONAL AGREEMENT CONCERNED, THE PREPARING ACTIVITY WILL TAKE APPROPRIATE RECONCILIATION ACTION THROUGH INTERNATIONAL STANDARDIZATION CHANNELS, INCLUDING DEPARTMENTAL STANDARDIZATION OFFICES, IF REQUIRED.

(A) IN THE EVENT OF A CONFLICT BETWEEN THE TEXT OF THIS STANDARD AND THE REFERENCES CITED HEREIN, THE TEXT OF THIS STANDARD SHALL TAKE PRECEDENCE.

(A) REFERENCED GOVERNMENT (OR NON-GOVERNMENT) DOCUMENTS OF THE ISSUE LISTED IN THAT ISSUE OF THE DEPARTMENT OF DEFENSE INDEX OF SPECIFICATIONS AND STANDARDS (DoDISS) SPECIFIED INTHE SOLICITATION FORM A PART OF THIS STANDARD TO THE EXTENT SPECIFIED HEREIN.

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APPROVED 30 NOV 73 REVISED (A) FOR CHANGES SHEETS 1 & 2

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AMSC N/A

P.A. NAVY — AS Other Cust USAF - 11 ARMY - AV	INTER-NATIONAL INTEREST *	TITLE: WRENCHING ELEMENT, EXTERNAL HEXAGON, FOR THREADED FASTENERS	MILITARY STANDARD MS21132
PROCUREMENT SPECIFICATION NONE		SUPERSEDES:	SHEET 2 OF 2

DD FORM 1 MAR 72 672-1 (Coordinated)

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